

MANTA

One continuous ribbon. One gesture.

Cut like a baguette so it can be printed at home and still hold an adult.



■ IDEA

MANTA is a single thick ribbon: floor, dished seat, cantilevered backrest, forward-curved crown, tail-brace, back to the floor — no separate legs, no frame, no visible assembly. The name is literal: forked feet read as fins, the tail-brace as a fluke. The ocean makes most of the oxygen we breathe, and the manta ray is one of its health indicators — so the surface carries the story too: flutes that follow the ribbon's own motion, bold at the floor and easing toward the crown, capped near the seat and back so they stay tactile without ever pressing into a bone.

■ BIOMIMICRY AS STRUCTURE, NOT ORNAMENT

Every centimetre of the section is set by a biological rule, and each is load-bearing:

- **Wolff's law** — thickness follows the bending moment: 84 mm at the lumbar knot, 50 mm at the knee, a 32 mm blade at the crown.
- **Murray's law** — the lumbar, the one true junction, is the cube-root-sum of seat, backrest and tail; every transition filleted.
- **Spiral fibre** — the flutes follow the stress path and stiffen by corrugation; the section twists up to 20° through the backrest.
- **Shell curvature** — the closed superelliptical section and dished seat carry load in membrane action.
- **Fiddlehead scroll** — the forward-curved crown closes the backrest into a tube, not an open cantilever.
- **Buttress roots** — each foot forks into two splayed prongs.

■ PRINTING

41 segments, each within 212 × 220 × 250 mm and pre-oriented on its own cut face to print **without supports** — every gram becomes chair, none scrapped as support waste. PETG, 0.2 mm layers, 5 perimeters, 35 % gyroid, solid pins. Finished weight ~16 kg, ~180 h batch print time, ~90 € in filament.

■ ASSEMBLY — SCREW-FREE

The ribbon is sliced **crosswise**, so every joint is a full solid cross-section (7,000–17,000 mm²), never a thin plate edge: a self-centring conical peg (Ø27) plus epoxy. Wide segments also split along the centre-line, seam hidden in a flute, reading as a keel. The two joints beside the lumbar get an added solid Ø11 pin. No screws, no brackets. Every peg is checked by virtual assembly to seat in its neighbour.

■ STABILITY AND LOAD

First-order check (120 kg × 1.8 dynamic × 2.0 safety), live from the width/thickness tables: every member passes; glued full-section joint 26×, weakest glued check 2.1×; with no glue at all, peg and pin still carry 0.9×. Tipping 0.56 sideways, 0.55–0.77 rearward; safe static load ~250 kg. Seat 429 × 364 mm, 428 mm high, 5° tilt, convex lumbar — 8/8 ergonomics checks pass, verified by script.

~250 kg

safe static load

26×

glued full-section joint

~16 kg

finished weight

~90 €

filament cost

■ OPEN DESIGN — NOTHING REPEATS

Nothing in nature repeats, so no two parts of this chair do either: all 41 segments are geometrically unique (script-verified, no duplicates). One parametric script: change the tables and it regenerates a leaner variant with the same joints, auto-checked against the same load case. A ready 90 kg, 40% lighter variant is included. Source, anonymised for review: anonymous.4open.science/r/manta-chair-23D4